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1- INTRODUCTION

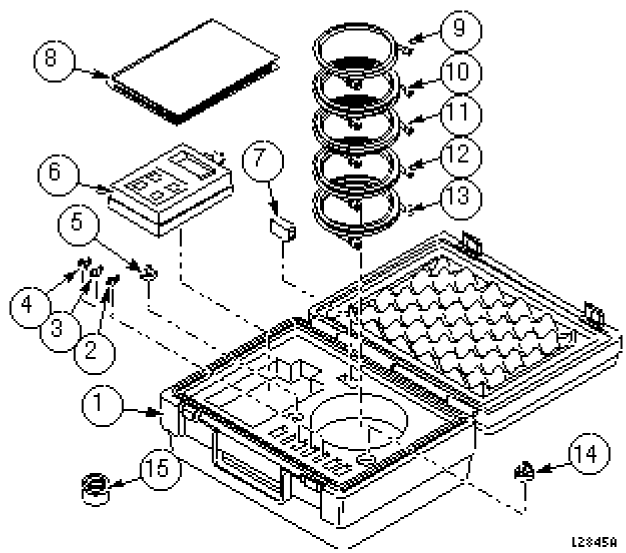
This procedure describes how to turn on the fiber optic drivers in a Signa system. Software output control is provided for the HFBR-1522 drivers (plastic cable) and the HFBR-1414 drivers (glass cable) on the TYME II card, the HFBR-1522s (plastic cable) on the PAC and the SRI, and the HFBR-1522 drivers (plastic cable) on the MDS peripherals. This procedure also describes how to measure the light output of the Fiber Optic Photoplethysmograph Probe.

1-1 Tools Required

Fiber Optic Light Meter Kit (46-317830G1). See items in Table 1-1 and Illustration 1-1.

TABLE 1-1
TOOLS AND INSTRUMENTS REQUIRED

ITEM	DESCRIPTION	PART NUMBER	QUANTITY
1	Case, Foam, Label	46-317830G50	1
2	SMA (Female - Female) Simplex Bulkhead Connector	46-287713P1	1
3	STC (Female - Female) Simplex Bulkhead Connector	46-317863P2	1
4	CFC (Female - Female) Simplex Bulkhead Connector	46-317863P1	1
5	Fiber Optic Adapter Assembly For FOT - 21 Meter	Supplied With Item 6	1
6	Fiber Optic Light Meter, 400-1014NM	46-317827P1	1
7	Spare +9v Battery	Supplied With Item 6	1
8	FOT - 21 Fiber Optic Light Meter Instruction Manual	Supplied With Item 6	1
9	ST Male - CFC Male (1 FT. Long)	46-317849P1	1
10	ST Male - ST Male (1 Meter Long)	46-307584P3	1
11	ST Male - SMA Male (1 Meter Long)	46-317847P1	1
12	ST Male - HP Male (1/2 Meter Long)	46-317848P1	1
13	HP Simplex Male Latching - Simplex Male Latching (1/2 Meter Long)	46-307564P4	1
14	HP Simplex/Duplex, Latching/Nonlatching Adapter For FOT-21 Meter	46-320033G1	1
15	Fiber Optic Adapter Assembly For F.O. Plethys. Probe	46-320710G1	1



FIBER OPTIC MEASUREMENT KIT CONFIG.
ILLUSTRATION 1-1

2- TYME-II, SRI, PAC, AND MDS LINK FIBER OPTIC CHECKS

WARNING!

POSSIBLE EYE INJURY! OBSERVING THE TRANSMITTER OUTPUT POWER UNDER MAGNIFICATION MAY CAUSE INJURY TO THE EYE. WHEN VIEWING WITH THE UNAIDED EYE, OR NORMAL EYE GLASSES, THE INFRARED OUTPUT IS RADIOLOGICALLY SAFE. HOWEVER, WHEN VIEWED UNDER MAGNIFICATION, PRECAUTION SHOULD BE TAKEN TO AVOID EXCEEDING THE LIMITS RECOMMENDED IN ANSI Z136.1 - 1981.

The Fiber Optic Test can be selected from the Diagnostics menu. The drivers on the TYME II card remain on until the Fiber Optic Test is canceled. The drivers on the MDS link remain on until the test is canceled, or a break occurs in the link upstream from the device being tested. The sequence of the boards on the MDS link is TYME II -> GAP (or GIP for SGD) -> CMB (or SSM for RF/PENII & RF/PDU) -> TYME II. So, for example, to test the link between the SSM and the TYME II, the Fiber Optic Test must be running and the links between the TYME II, GIP, and SSM must be intact. If a break does occur upstream from the device under test, the Driver On condition will not propagate through the MDS link.

2-1 Procedure

1. To turn on the HFBR-1522 drivers on the TYME card and the HFBR-1522 drivers on the MDS peripherals, select **[Diagnostics]** from the Service Desktop, **[Diagnostic Main Menu]**, and **[Start…]**. In the Diagnostics window, select **[IPG], [Manual]**, **[Fiber Optics from TPS]**, and finally **[Run Diags]**.

2. To turn on the HFBR-1522s (plastic cable) on the PAC and the SRI, , select **[Diagnostics]** from the Service Desktop, **[Diagnostic Main Menu]**, and **[Start…]**. In the Diagnostics window, select **[IPG], [Manual]**, **[Fiber Optics to TPS]**, and finally **[Run Diags]**.

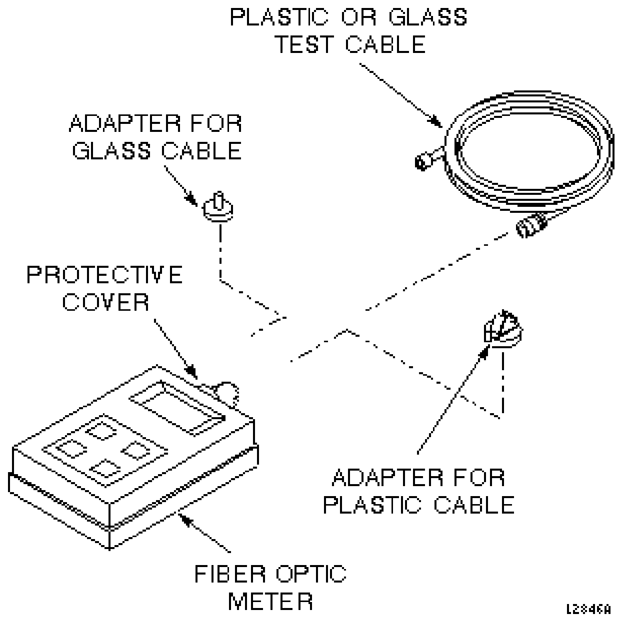
Note

After selecting the fiber optics test, a note is displayed at the bottom of the “MAN” (IPG manual diags) menu box instructing you to use the light instrument to measure the intensity in the fiber optic cable.

Note

Fiber optic test resets the TPS/IPG and displays the message “Running IPG Manual Diags” in the Diags Status box.

3. Remove protective cover and install the appropriate adapter (46-320033G1 for plastic cables; FOA-32 for glass cables) onto the fiber optic meter. Plug test cable (1/2 m for plastic (46-307564P4), 1 m for glass (46-307584P3)) onto the adapter. See Illustration 2-1.



FIBER OPTIC MEASUREMENT CONFIGURATION
ILLUSTRATION 2-1

4. Use the **[on/off]** button to turn on the EXFO fiber optic meter. Make sure the meter is on 660 calibration for plastic links and 820 calibration for glass links. Use the **[λ Select]** button to alternate between wavelength values displayed on the meter LCD display. Use the **[dBm/W]** button to select dBm unit scale on the meter LCD display.

CAUTION

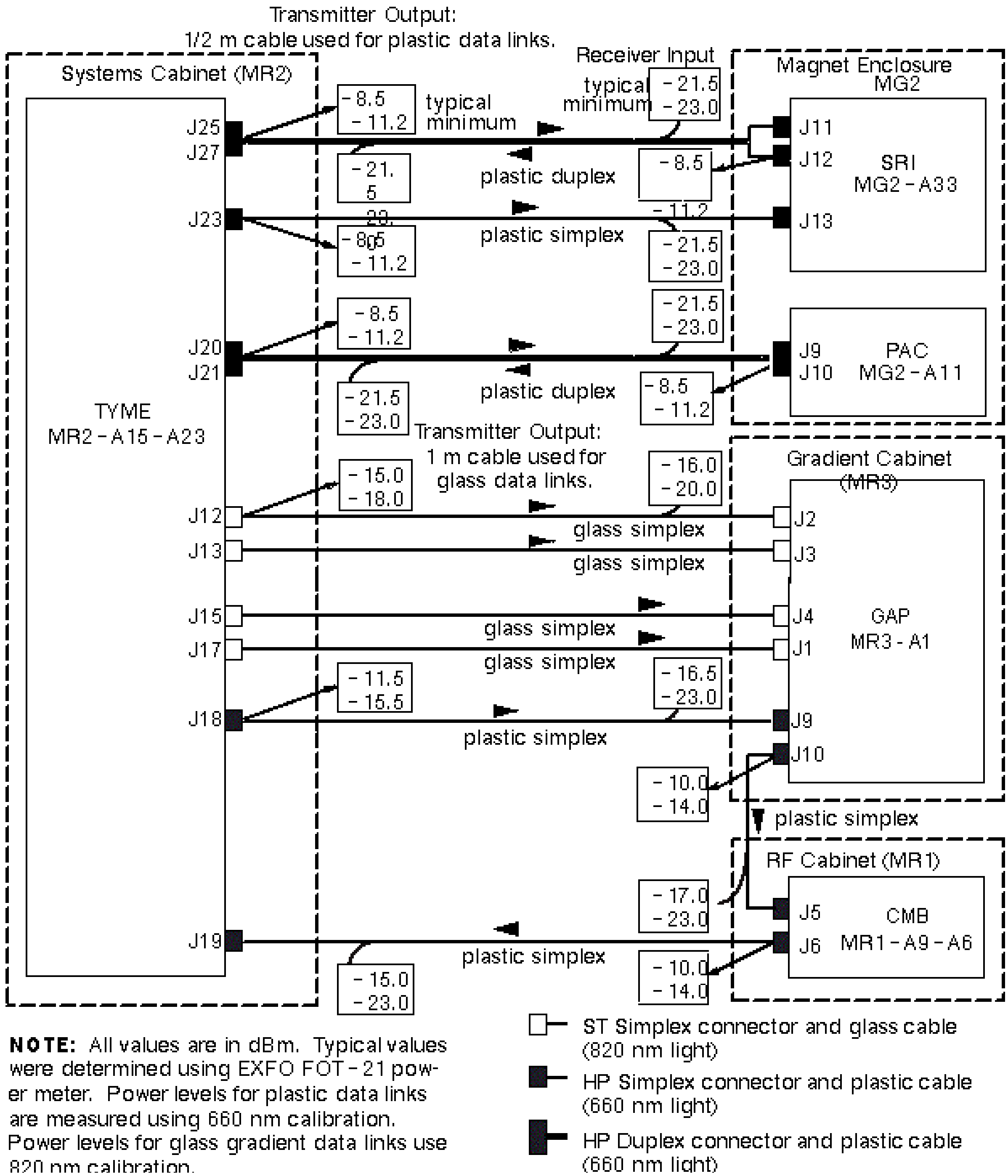
Cable damage possibility. Disconnecting the fiber optic cables by pulling on the cable can damage the cable/connector. Use the connector to disconnect cables.

5. Disconnect system cable from transmitter (gray fiber optic connector) under test. Connect the test cable into the transmitter under test.

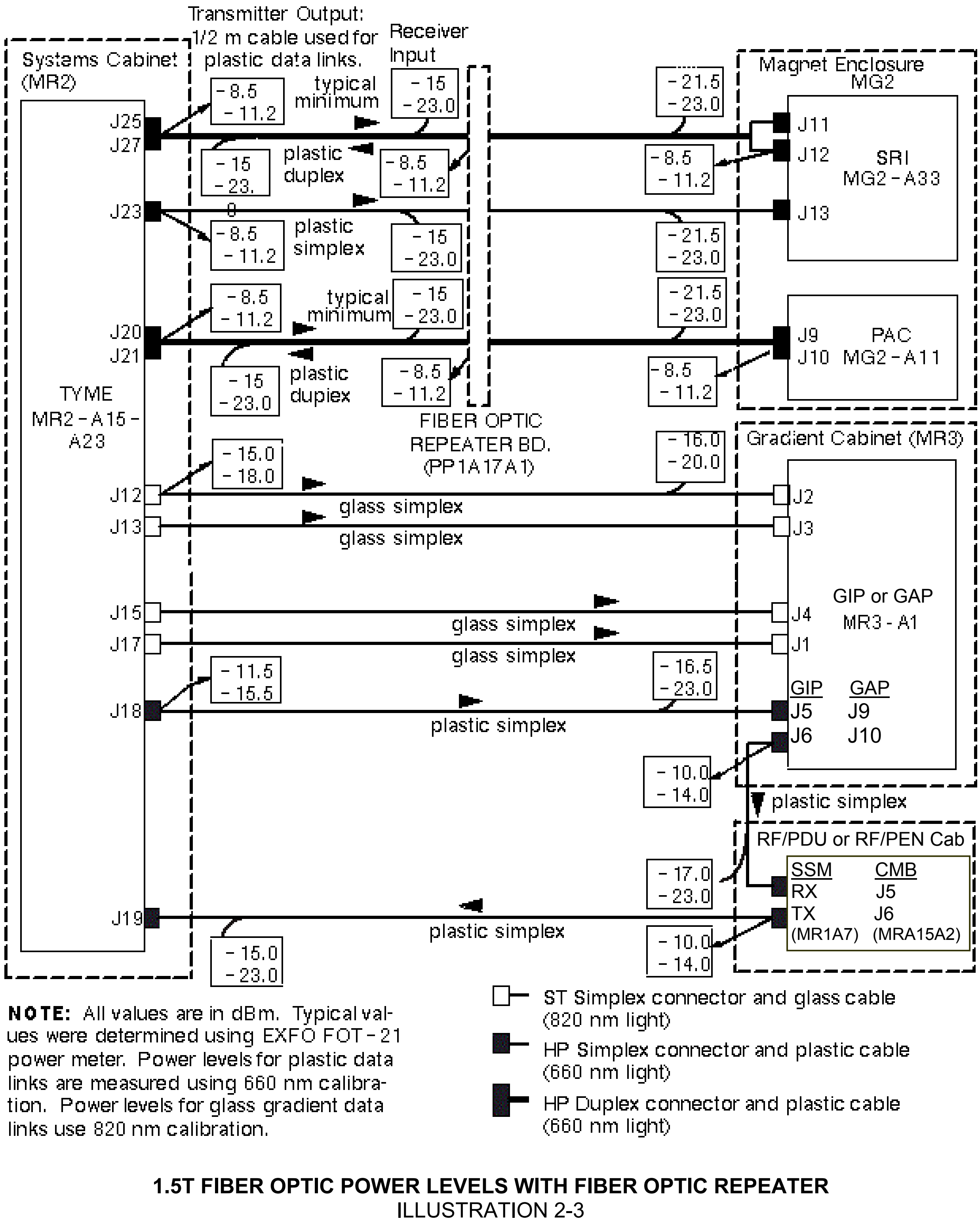
Note

The fiber optic connectors are a latching type connector. For the plastic cable connectors, press on the latch to release. For the glass cable connectors, press in on outer connector shell and rotate a quarter turn.

6. Measure transmitter output through test cable with EXFO fiber optic meter. Compare this value with specification. See Illustration 2-2 or Illustration 2-3 (with Fiber Optic Repeater). If value is out of spec, there is a problem. It could be a power supply problem, transmitter problem, or functional problem with board. Do not proceed until this value is within spec.



1.5T FIBER OPTIC POWER LEVELS
ILLUSTRATION 2-2



NOTE: All values are in dBm. Typical values were determined using EXFO FOT-21 power meter. Power levels for plastic data links are measured using 660 nm calibration. Power levels for glass gradient data links use 820 nm calibration.

1.5T FIBER OPTIC POWER LEVELS WITH FIBER OPTIC REPEATER
ILLUSTRATION 2-3

Note

To troubleshoot the fiber optic communications link between the Bit3 (VME I/F) Board in the Systems Cabinet and the Bit3 (VME I/F) Board in the Host Computer, refer to procedure “Bit 3 Communication Troubleshooting”. *This proprietary procedure is available for GE use, and to sites with a valid Advanced Service Package Limited License.*

Note

The fiber optic connectors are a latching type connector. For the plastic cable connectors, press on the latch to release. For the glass cable connectors, press on the outer connector shell and rotate a quarter turn.

7. Disconnect test cable from the transmitter under test and reconnect system cable to transmitter.

8. Disconnect test cable from EXFO fiber optic meter.

9. Disconnect far end of system cable from receiver (blue fiber optic connector; gray for some) and connect system cable to meter adapter.

10. Measure optical output from system cable. Compare this value with specification. If this value is out of spec and the transmitter output was within spec, there is a cable routing issue or a damaged cable. Check cable for kinks, tight radius bends, damaged or poorly polished connectors, etc.

11. Repeat this procedure for each link that requires testing.

2-2 Test Specifications

The following data characterize fiber optic power levels in a Signa system. Typical data were obtained from a limited sample of systems. Typical values may cover a wide range due to the effects of cable routing, connecting, and cable length which may vary from system to system. Minimum values, however, are absolute, and will be the same for fixed sites, mobiles, and T/Rs.

Plastic Data Links (AMP fiber 501232-5 or 501336-1 with HP versatile link connectors and 660 nm HFBR-1522/ HFBR-2522 transmitter/receiver pair):

TYME II J25 to SRI J11; TYMEII J23 to SRI J13; SRI J12 to TYME II J27; TYME II J20 to PAC J9; PAC J10 to TYME II J21:

- Nonmodulated HFBR-1522 transmitter output through 1/2 m test cable:
  Typical: –8.5 dBm
  Minimum: –11.2 dBm
- Nonmodulated HFBR-1522 transmitter output through 147 ft cable:
  Typical: –21.5 dBm
  Minimum: –23.0 dBm

GAP J10 to CMB J5 <u>or</u> GIP J6 to SSM RX:

- Nonmodulated HFBR-1522 transmitter output through 1/2 m test cable:
  Typical: –10.0 dBm
  Minimum: –14.0 dBm
- Nonmodulated HFBR-1522 transmitter output through 82 ft cable:
  Typical: –17.0 dBm
  Minimum: –23.0 dBm

TYME J18 to GAP J9 <u>or</u> GIP J5:

- Nonmodulated HFBR-1522 transmitter output through 1/2 m test cable:
  Typical: –11.5 dBm
  Minimum: –15.5 dBm
- Nonmodulated HFBR-1522 transmitter output through 46 ft cable:
  Typical: –16.5 dBm
  Minimum: –23.0 dBm

CMB J6 <u>or</u> SSM TX to TYME II J19:

- Nonmodulated HFBR-1522 transmitter output through 1/2 m test cable:
  Typical: –10.0 dBm
  Minimum: –14.0 dBm
- Nonmodulated HFBR-1522 transmitter output through 46 ft cable:
  Typical: –15.0 dBm
  Minimum: –23.0 dBm

Fiber Optic Repeater (FOR) Data Links:

TYME II J25 to FOR J9; TYME II J23 to FOR J11; FOR J10 to TYME II J27; TYME II J20 to FOR J7; FOR J8 to TYME II J21:

- Nonmodulated HFBR-1522/1532 transmitter output through 1/2 m test cable:
  Typical: –8.5 dBm
  Minimum: –11.2 dBm
- Nonmodulated HFBR-1522/1532 transmitter output through 60 ft cable:
  Typical: –15 dBm
  Minimum: –23.0 dBm

FOR J5 to SRI J13; FOR J3 to SRI J11; FOR J1 to PAC J9; PAC J10 to FOR J2; SRI J12 to FOR J4:

- Nonmodulated HFBR-1522/1532 transmitter output through 1/2 m test cable:
  Typical: –8.5 dBm
  Minimum: –11.2 dBm
- Nonmodulated HFBR-1522/1532 transmitter output through 90 ft cable:
  Typical: –17.5 dBm
  Minimum: –23.0 dBm

Gradient (glass) Data Links: (AMP fiber 502083-1 with ST type connectors and 820 nm HFBR-1414/HFBR-2416 transmitter/receiver pair)

TYME II J12 to GIP or GAP J2; TYME II J13 to GIP or GAP J3; TYME II J15 to GIP or GAP J4; TYME II J18 to GIP or GAP J1:

- Nonmodulated HFBR-1414 transmitter output through 1 m test cable:
  Typical: –15 dBm
  Minimum: –18.0 dBm
- Nonmodulated HFBR-2416 transmitter output through 46 ft glass cable:
  Typical: –16 dBm
  Minimum: –20.0 dBm

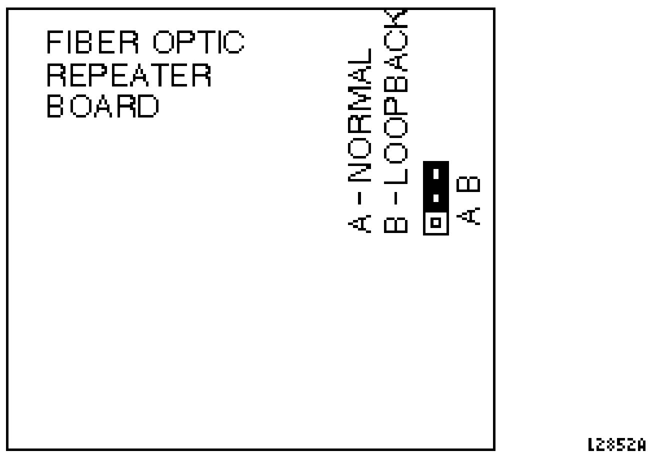
3- FIBER OPTIC PHOTOPLETHYSMOGRAPH PROBE CHECKOUT

Refer to “Peripheral Pulse Probe Check” procedure.

4- FIBER OPTIC REPEATER LOOPBACK TEST

For systems with a Fiber Optic Repeater at the Penetration Panel, the Fiber Optic Repeater test is provided to test lines between the System Cabinet and the Fiber Optic Repeater Board (except SRI Reset). When the test runs, the IPG sends a data pattern out to the Fiber Optic Repeater Board, and verifies that the exact same data is looped back to the IPG. Any mismatches indicate that either the loopback jumper is not set, or there is a fault in the lines between the IPG and the Fiber Optic Repeater Board.

1. Set loopback jumper in the Fiber Optic Repeater Board (PP1A17A1) to Position "B" (Loopback). See Illustration 4-1.



FIBER OPTIC REPEATER BOARD LOOPBACK JUMPER
ILLUSTRATION 4-1

2. At the Operator work space select **[Tools Icon]**, **[Diagnostics]**, **[Diagnostic Main Menu]**, and **[Start…]**. In the Diagnostics window, select **[IPG], [Manual]**, **[FiberOptic Repeater]**, and then **[Run Diags]**.

An error is logged if any of the following conditions occur:

- A fiber optic line between the Tyme II board and the Fiber Optic Repeater is defective (this test does not include the SRI Reset fiber optic).
- The system does not have a Fiber Optic Repeater.
- The loopback jumper at the Fiber Optic Repeater is NOT in position "B".
- The Fiber Optic Repeater board is defective.

5- SYSTEM RESTORATION

1. Turn off the fiber optic meter. Disconnect all kit components used in measurement scheme and return to kit. See Illustration 1-1.

2. Restore all system fiber optic cables and components to original configuration.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	June 1, 1998	J. Saperstein	Initial Conversion from Toolbook to Word.
1	Jan. 5, 1999	K. Keshena	Add Illustrations 2-2 and 2-3. Updated styles.
2	Oct. 18, 1999	M. Keber	Corrected menu names, updated Illustration 2-3 and test spec section for SGD and RF/PDU hardware, replaced Peripheral Pulse Probe check section with reference to existing documented procedure.